

Telecom Database Normalization

A telecom company stores customer subscription data in one table.

customer_id	customer_name	phone_number	plan_name	plan_price	services	devices	tower_id	tower_location
C001	Aung Aung	09123456789	Premium	30000	Voice, SMS, Data	iPhone, Router	T01	Yangon
C002	Su Su	09987654321	Basic	10000	Voice, SMS	Samsung	T02	Mandalay
C003	Kyaw Kyaw	09777777777	Premium	30000	Voice, SMS, Data	Router	T01	Yangon

Q1. What normalization problems exist in the original telecom table?

Problems in the original table:

1. Repeating groups / multivalued attributes

- Services: Voice, SMS, Data
- Devices: iPhone, Router

2. Data redundancy

- Plan details repeated for many customers
- Tower location repeated
- Customer details repeated when multiple services/devices exist

3. Update anomalies

- Changing plan price requires multiple updates
- Tower location changes require repeated modifications

4. Insertion anomalies

- Cannot add a new plan without customer data

5. Deletion anomalies

- Deleting customer may remove plan or tower information

Main issue: The table is not in First Normal Form (1NF) because of multivalued fields.

Q2. Convert the telecom table into 1NF

customer_id	customer_name	phone_number	plan_name	plan_price	service	device	tower_id	tower_location
C001	Aung Aung	09123456789	Premium	30000	Voice	iPhone	T01	Yangon
C001	Aung Aung	09123456789	Premium	30000	Voice	Router	T01	Yangon
C001	Aung Aung	09123456789	Premium	30000	SMS	iPhone	T01	Yangon
C001	Aung Aung	09123456789	Premium	30000	SMS	Router	T01	Yangon
C001	Aung Aung	09123456789	Premium	30000	Data	iPhone	T01	Yangon
C001	Aung Aung	09123456789	Premium	30000	Data	Router	T01	Yangon
C002	Su Su	09987654321	Basic	10000	Voice	Samsung	T02	Mandalay
C002	Su Su	09987654321	Basic	10000	SMS	Samsung	T02	Mandalay
C003	Kyaw Kyaw	0977777777	Premium	30000	Voice	Router	T01	Yangon
C003	Kyaw Kyaw	0977777777	Premium	30000	SMS	Router	T01	Yangon
C003	Kyaw Kyaw	0977777777	Premium	30000	Data	Router	T01	Yangon

Q3. What is the main issue after converting to 1NF?

Main issue: Partial dependency

Example composite key:
(customer_id, service, device)

Problems:

- customer_name depends only on customer_id
- plan_name depends only on customer_id
- tower_id depends only on customer_id

Conclusion: The table is not in Second Normal Form (2NF).

Q4. Convert the 1NF table into 2NF

Customer Table

customer_id	customer_name	phone_number	plan_name	tower_id
C001	Aung Aung	09123456789	Premium	T01
C002	Su Su	09987654321	Basic	T02
C003	Kyaw Kyaw	09777777777	Premium	T01

Customer_Service Table

customer_id	service
C001	Voice
C001	SMS
C001	Data
C002	Voice
C002	SMS
C003	Voice
C003	SMS
C003	Data

Customer_Device Table

customer_id	device
C001	iPhone
C001	Router
C002	Samsung
C003	Router

Q5. What issue still exists after 2NF?

Main issue: Transitive dependency

Examples:

- plan_name → plan_price
- tower_id → tower_location

Problem: Non-key attributes depend on other non-key attributes.

Q6. Convert the 2NF design into 3NF

Customer Table

customer_id	customer_name	phone_number	plan_id	tower_id
C001	Aung Aung	09123456789	P01	T01
C002	Su Su	09987654321	P02	T02
C003	Kyaw Kyaw	09777777777	P01	T01

Plan Table

plan_id	plan_name	plan_price
P01	Premium	30000
P02	Basic	10000

Tower Table

tower_id	tower_location
T01	Yangon
T02	Mandalay

Customer_Service Table

customer_id	service
C001	Voice
C001	SMS
C001	Data
C002	Voice
C002	SMS
C003	Voice
C003	SMS
C003	Data

Customer_Device Table

customer_id	device
C001	iPhone
C001	Router

customer_id	device
C002	Samsung
C003	Router

Q7. Is the telecom database now in BCNF?

Yes.

Because:

- customer_id determines customer information
- plan_id determines plan details
- tower_id determines tower details

Conclusion: The design satisfies BCNF.

Q8. What 4NF issue can exist in telecom data?

Issue: Multivalued dependencies

Example:

A customer may independently have:

- Multiple services
- Multiple devices

Solution:

Separate tables:

- Customer_Service
- Customer_Device

Q9. What is a possible 5NF issue in telecom data?

Issue: Join dependency

Example:

Combining:

- Customers
- Services

- Devices

may generate false combinations if stored together.

Solution: Use separate relationship tables and join only when needed.

Q10. What is the final normalized telecom database design?

Final Database Structure:

Customer

- customer_id (PK)
- customer_name
- phone_number
- plan_id (FK)
- tower_id (FK)

Plan

- plan_id (PK)
- plan_name
- plan_price

Tower

- tower_id (PK)
- tower_location

Customer_Service

- customer_id (FK)
- service

Customer_Device

- customer_id (FK)
- device